EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

GS-1 (DYNAMIC)

REVISION

MODULE - 60

ECOLOGY & ENVIRONMENT-12

Why is Dugong in the news recently?

- The estimated population of Dugongs in Tamil Nadu has been put at 270. The survey was undertaken by the Wildlife Institute of India, (WII), in collaboration with the Forest Department along the Tamil Nadu coast.
- There are 158 Dugongs in Palk Bay and 112 Dugongs in Gulf of Mannar. As per the Tamil Nadu government officials, the Dugong population in Tamil Nadu is stable and doing well.
- Taken together, Palk Bay and Gulf of Mannar now represent the largest viable Dugong population in India. This is largely attributed to the implementation of the Dugong Recovery Program by WII, supported through CAMPA, along with the proactive and sustained conservation efforts of the Forest Department, including the community engagement.
- The report also highlights the notification of the dugong Conservation Reserve in Palk Bay, covering 448 square km of its core habitat by the government, as a significant milestone towards long-term protection of the species.

What do you know about the recent mandate given with regard to 20% ethanol blended petrol?

Octane is like a fuel's self-control under pressure - higher RON means fuel stays stable under high compression. Ethanol naturally has a high-octane value at around 108 RON. Blending 20% ethanol in petrol increases knock resistance.

- Government has mandated the sale of petrol with up to 20% ethanol and a minimum Research Octane Number (RON) of 95 across all States and the Union Territories from April 1, 2026.
- The Central Government can allow exceptions in special situations, for specific regions and for a limited time.
- Ethanol is made from sugarcane, maize or grain. Most vehicles manufactured in India from 2023-2025 onward are designed to run on E20 and no major issues are expected.
- Older vehicles may however see a slight drop in mileage (3-7%) along with the wearing out of rubber/plastic components. The insistence on minimum RON 95 is to prevent engine damage.
- RON or Research Octane Number is a measure of a fuel's resistance to engine knocking (Pre-ignition). Knocking happens when fuel burns unevenly inside the engine, causing pinging sound, loss of power, and possible engine damage over time. The higher the RON, the more resistant the fuel is to knocking.

This is a snapshot on Ethanol supply to OMCs and Ethanol supply by feedstock

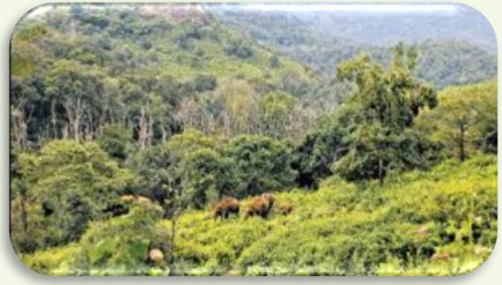
• Ethanol supply to oil marketing companies

• Ethanol supply by feedstock (in crore litres)

Supply Year	C-Heavy Molasses	B-Heavy Molasses	Cane Juice	Maize	Surplus FCI Rice	Damaged Foodgrain
2013-14	38	0	0	0	0	0
2014-15	67	0	0	0	0	0
2015-16	111	0	0	0	0	0
2016-17	67	0	0	0	0	0
2017-18	151	0	0	0	0	0
2018-19	146	33	1	0	0	10
2019-20	74	68	15	0	0	16
2020-21	39	183	39	0	2	39
2021-22	11	265	85	0	49	24
2022-23	6	235	128	32	74	32
2023-24	58	149	64	286	0	116
2024-25	17	139	165	498	140	80
2025-26*	14	113	165	479	233	54

*CONTRACTED QUANTITY TILL NOW. SUPPLY YEARS ARE DEC-NOV TILL 2021-22; DEC-OCT FOR 2022-23 AND NOV-OCT FROM 2023-24.

What do you know about the recent rules notified by the Union Ministry of Environment, Forests and Climate Change (MoEFCC) ?



- Recently, MoEFCC notified that assisted natural regeneration, afforestation and plantations carried out by government or non-government entities are to be treated as forestry activities. Such projects will no longer require payment of NPV and the creation of compensatory afforestation plots.
- Environmentalists questioned the technicality of the fresh guidelines and argues that when the Union Government leases out forest lands, it inevitably counts as forest diversion because the lessee or the user agency has the right to hold and use the land on certain conditions for a specified period.
- Raising of plantations or assisted natural regeneration, which involves benefit sharing with the lessee, is a commercial activity profitable to the user agency (the lessee), similar to a mining lease for which the lessee is bound to pay NPV and undertake compensatory afforestation. Exempting such user agencies from these compliances is greenwashing the diversion, as per environmentalists.

As 60 countries ratified the treaty, it came into force. India signed but, yet to ratify.

HIGH SEAS TREATY HAS ENTERED INTO FORCE IN JANUARY 2026

- Biodiversity Beyond National Jurisdiction (BBNJ) or High Seas Treaty has entered into force recently, creating the first legally binding global framework to conserve and sustainably use the high seas.
- High seas refer to ocean areas beyond national jurisdiction, i.e., beyond 200 nautical miles. They cover nearly two-thirds of the global oceans and about half of the Earth's surface.
- It was adopted in 2023 by an intergovernmental conference on marine biodiversity of areas beyond national jurisdiction convened under the auspices of the United Nations.
- Till now, the biodiversity is governed mainly by General principles under UNCLOS with limited biodiversity specific regulation.

BIODIVERSITY OF AREAS BEYOND NATIONAL JURISDICTION (BBNJ) OR THE HIGH SEAS TREATY IS THE THIRD AGREEMENT UNDER UNCLOS

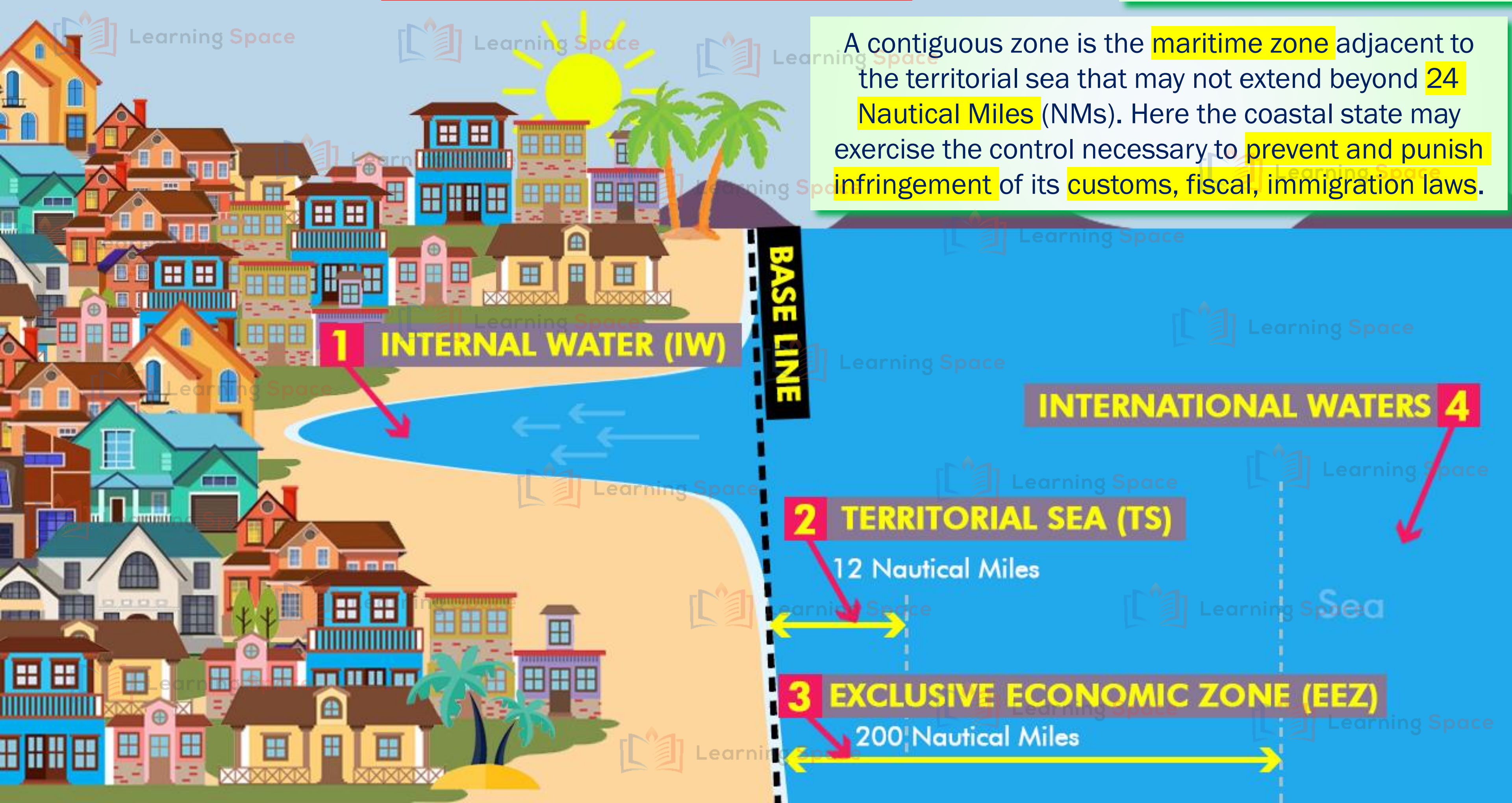
- It has three objectives:
  - ✓ Conserving marine biodiversity.
  - ✓ Ensuring equitable sharing of benefits from marine genetic resources.
  - ✓ Mandating environmental impact assessments for harmful activities.

- This is the third agreement to be approved under UNCLOS, after the 1994 and 1995 treaties, which established the International Seabed Authority and the Fish Stocks Agreement.
- Decisions will be taken via Conference Of Parties (COPs). It creates a clearinghouse mechanism for data, knowledge, monetary benefits and cooperation.
- It sets up a secretariat and a dedicated funding mechanism to support implementation.
- A scientific and technical body will be created, which will play a significant role in environmental impact assessment. It will be creating standards and guidelines for assessment procedures and helping countries with less capacity in carrying out assessments.

UNCLOS, 1982 IN A NUTSHELL!

1 Nautical Mile = 1.852 KM.

A contiguous zone is the maritime zone adjacent to the territorial sea that may not extend beyond 24 Nautical Miles (NMs). Here the coastal state may exercise the control necessary to prevent and punish infringement of its customs, fiscal, immigration laws.



MAJOR PROVISIONS OF THE HIGH SEAS TRADE TREATY (BBNJ)

1 MARINE PROTECTED AREAS (MPAs)

- It facilitates establishing marine protected areas to protect oceans from human activities through a three quarterly majority vote. The goal is to establish global network of MPAs. This is critical for 30% ocean protection by 2030.
- The objective is to implement international regulations to protect life in oceans beyond national jurisdiction through international cooperation.

2 ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Mandatory for significant harm activities. It applies to large scale fishing, deep sea exploration, infrastructure. It enhances precautionary and ecosystem-based approaches.

MAJOR PROVISIONS OF THE HIGH SEAS TREATY (BBNJ)

3 MARINE GENETIC RESOURCES AND BENEFIT SHARING

- It plans to equitably share benefits (financial and non-financial) from blue economy.
- Addresses developing country bio-prospecting concerns.

4 CAPACITY BUILDING AND TECHNOLOGY TRANSFER

- It supports developing and least developed countries.
- Focus is on marine research, data sharing, technology access, and scientific collaboration.

WHAT ARE THE CONCERNS EXPRESSED BY EXPERTS?

Learning Space

- The treaty does not regulate deep sea mining, which is under the International Seabed Authority (ISA).
- Monitoring and enforcement in vast Oceans is technically and financially difficult.
- The USA has signed but not ratified. The treaty relies more on cooperation raising concerns over compliance.
- There is no consensus on marine protected areas.
- The provision pertaining to marine genetic resources may be undermined by wealthier nations under-reporting their activities.
- Capacity building and technology transfers present another challenge for low and middle-income countries.
- Some of the provisions may conflict with regimes like the Convention on Biological Diversity.

As per critics, the implementation of this ambitious treaty is difficult because of overlapping maritime claims - for example, South China Sea.

COP30 ACKNOWLEDGED THAT MORE CLIMATE ACTION IS NEEDED BUT FAILED TO PROVIDE IT!

- The World Meteorological Organization said 2025 is on track to be the second or third warmest year on record, extending an alarming run of exceptional temperatures.
- The UNEP's Emissions Gap Report warned that the world is heading for a 2.8 degrees Celsius of warming and that global temperatures are likely to exceed 1.5 degrees Celsius above pre-industrial levels within the next decade. Overshooting that threshold will unleash spiraling, unchecked climate impacts.

After COP30, UN Secretary General Antonio Guterres summarized … "the gap between where we are and what science demands remains dangerously wide".

KEY OUTCOMES

- A new mechanism for international cooperation on a just transition.
- Tripling adaptation finance by 2035.
- A work program under Article 9 of the Paris Agreement to scrutinize finance provided by developed countries to help developing ones mitigate and adapt to climate change.

DEMANDS THAT COULD NOT BE AGREED ARE RELEASED AS GLOBAL MUTIRAO DECISION

- Several key demands of developing countries that could not make it onto the main agenda items for negotiation are released as Global Mutirao Decision (Mutirao means collective effort).
- These include four contentious issues:
  - ✓ Provision of climate finance by developed countries.
  - ✓ Responding to 1.5 degrees Celsius ambition and implementation gap.
  - ✓ Transparency arrangements under the Paris Agreement.
  - ✓ Addressing concerns over climate-related unilateral trade measures.

KEY OUTCOMES

1 ADAPTATION SUPPORT

DECISION

- Under Article 9.1 of the Paris Agreement, developed countries are legally entitled to provide finance for developing countries for climate action.
- International public adaptation finance amounted to only $26 billion in 2023, barely 1/12th of what is required.
- It was decided to triple adaptation finance by 2035. COP30 also adopted 59 indicators under Global Goal on Adaptation.

MAJOR GAPS

- The goal to triple was delayed to 2035 instead of 2030.
- The baseline was also undefined in the text.
- The 59 indicators adopted are known as Belem adaptation indicators. These are intended to form the backbone of the UAE framework for global climate resilience agreed in Dubai at COP29. The indicators package does not include indicators on means of implementation - mainly finance and technology transfer.

KEY OUTCOMES

2 CLIMATE FINANCE

DECISION

- Developing countries pushed for clearer accountability for the $300 billion annually by 2035 under NCQG.
- A two-year work program on Article 9 was established.

However, the only solace has been the launch of the Barbados Implementation Modalities (BIM), the first operational phase of the Fund for Responding to Loss and Damage (FRLD).

MAJOR GAPS

- Article 9(1) sets out developed countries' legal obligation to provide climate finance.
- The two-year work program on Article 9(1) also includes Article 9 as a whole - including Article 9(3), which calls for finance to be mobilized from a wide variety of sources. This risks crowding out rich countries accounting for delivering public finance under the $300 billion a year goal.
- With regard to Baku to Belem roadmap to 1.3T - Climate finance flows from all actors should be at least $1.3 trillion annually for developing countries by 2035 – was merely noted down.

KEY OUTCOMES

OTHER IMPORTANT ASPECTS ABOUT CLIMATE FINANCE

- Barbados Implementation Modalities (BIM), the first operational phase of the Fund for Responding to Loss and Damage (FRLD) has been launched. The BIM will accept funding proposals from developing countries also.
- In tune with Article 2(1)(c) of the Paris Agreement, which requires aligning finance flows with low emissions and climate-resilient development, a Veredas Dialogue, including at least one annual meeting and a high-level roundtable and the Xingu Finance Talks involving academia, international financial institutions and the private sector were agreed.

KEY OUTCOMES

3 JUST TRANSITION

DECISION

- In a significant win to the G77 (a coalition of 134 countries) and China, COP30 adopted a new Just Transition Mechanism (JTM).
- It is to implement just transition pathways by boosting international cooperation, providing technical assistance and building capacity.
- It recognizes need to scale up new and additional grant-based and highly concessional finance, including non-debt instruments.

MAJOR GAPS

- The decision contains no reference to climate-related trade restrictive unilateral measures.
- There is no explicit mention of the obligation of developed countries to provide the means of implementation that would enable developing countries to accelerate their domestic transitions.
- China objected to any mentioning of critical minerals, as EU and UK pushed for diversifying supply chains and sharing benefits fairly.

KEY OUTCOMES

Learning Space

4 OTHER IMPORTANT DECISIONS

1. The COP30 "takes note" of the Article 6 based non-market approaches platform and encourages countries to make greater use of its mitigation measures. Article 6 provides a framework for international cooperation on climate action through market and non-market-based mechanisms, allows countries to cooperate voluntarily to meet their climate goals or NDCs.
2. Under Article 6(2) of the Paris Agreement, which governs direct carbon trading between countries, talks are centered on bolstering transparency and tightening reporting and review.
3. There is no clear unanimity about Article 6.4, which establishes a UN-supervised global carbon market.
4. Another sticking point was the transition of old Clean Development Mechanism (CDM) projects into new mechanism.

KEY OUTCOMES

4 OTHER IMPORTANT DECISIONS

5. With regard to Carbon Border Adjustment Mechanism (CBAM), developing countries argued that CBAM shifts decarbonization costs onto poorer economies, contradicts UNFCCC agenda and undermines multilateral cooperation.
6. They pointed to Article 3.5 of the Convention, which warns that climate measures, including unilateral ones, must not amount to arbitrary discrimination or disguised restrictions on trade.
7. Developing countries say that CBAM is an extraterritorial policy that exacerbates economic and commercial pressures.

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